

Koide's 2/3 as a Generation-Coherence Structure: an Event Density Re-description, a Toy Doublet-Grid Map, and a Neutrino Prediction

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Abstract

The Koide relation — that the three charged-lepton masses satisfy $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ to one part in 10^5 — is a genuine, complete-data regularity (there are exactly three generations; none is omitted). We record three things. First, a compact restatement: writing each fermion family in Brannen form places the three generations 120° apart on a circle, and the entire Koide phenomenology reduces to the one line $\mathbf{Q} = \mathbf{1}/\mathbf{3} + \mathbf{w}^2/\mathbf{6}$, where the “wobble depth” w is $\sqrt{2}$ for the charged leptons (the value that yields 2/3) and larger for the quark families. Second, an Event Density re-description: since mass is rest-frame bandwidth content ([Paper_113](#)) and ED's carrier is $P = \sqrt{b} e^{i\pi_K}$ (with π_K the polarity phase), the $\sqrt{m}$ in Koide's formula plays the role of the ED amplitude of a mass-mode — treating the three generations as coherent amplitude-channels, an organizing choice rather than a corpus result — so Q reads as a coherence ratio among three generation-amplitudes. This is a shape-match, not a derivation. Third, two toy maps that place the wobble over the fermions' real quantum numbers (electric charge; then weak isospin $\times$ hypercharge); both are underdetermined by the three charged data points, and both extrapolate to the same single prediction — a neutrino Koide value $\mathbf{Q} \approx \mathbf{0.77}$ — which is untestable until neutrino masses sharpen. We are explicit throughout that ED re-describes and toy-maps this structure; it does not derive it.

1. The relation, and the honest weight it carries

Koide (1981) observed that the three charged-lepton pole masses satisfy

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = 0.666659 \approx \frac{2}{3},$$

precise to ~ 1 part in 10^5 , and *tightening* toward 2/3 as the τ mass has been remeasured over four decades. Two honest points frame everything below.

This is a complete-data regularity, not a small sample. There are exactly three charged-lepton generations; no fourth exists to measure, and none was left out. So Koide is the *entire* dataset producing one dimensionless number that lands on a simple fraction. That is why it is

taken seriously and is not dismissible as a coincidence among many trials — there are no other trials.

It is clean only for the charged leptons. The up- and down-quark families give $Q \approx 0.85$ and 0.73 , but quark masses are scheme-dependent (running values), so those numbers are soft. Neutrino masses are essentially unknown. The rigid, remarkable fact is the charged-lepton $2/3$.

2. The geometry, and one law behind every family

Write each family in the **Brannen (2005) form**:

$$\sqrt{m_k} = M \left(1 + w \cos\left(\delta + \frac{2\pi k}{3}\right) \right), \quad k = 0, 1, 2,$$

three points **120° apart on a circle** (a Z_3 / democratic arrangement) with overall scale M , wobble depth w , and phase offset δ . Each dot's horizontal projection is a $\sqrt{m}$ (Figure 1). Because the three cosines sum to zero and their squares sum to $3/2$, one gets $\Sigma\sqrt{m} = 3M$ and $\Sigma m = M^2(3 + 3w^2/2)$, so the whole Koide quantity collapses to

$$Q = \frac{1}{3} + \frac{w^2}{6}.$$

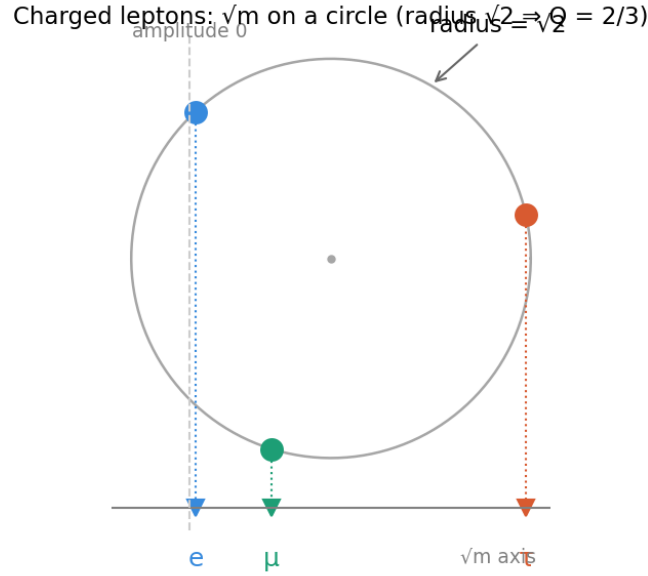


Figure 1: The three charged leptons as $\sqrt{m}$ on a circle of radius $\sqrt{2}$ (Brannen form). Each dot's horizontal position is a $\sqrt{m}$; the radius $\sqrt{2}$ is exactly the value that makes $Q = 2/3$, equivalently the $\sqrt{m}$ vector sitting at 45° to the democratic $(1,1,1)$ direction ($\cos = 1/\sqrt{2}$).

So **Q depends only on the wobble depth w** , not on δ or M . The families differ only in w : charged leptons at $w = \sqrt{2}$ ($\Rightarrow Q = 2/3$ exactly — the $\sqrt{m}$ vector at 45° to the $(1,1,1)$ diagonal, $\cos = 1/\sqrt{2}$), down-quarks at $w \approx 1.55$, up-quarks at $w \approx 1.76$ (Figure 2). The phase δ rotates the triangle and sets the *ratios* among the three masses (leptons: $1 : 207 : 3477$, the real values)

without touching Q . For the leptons $\delta \approx 0.222 \approx 2/9$ rad — a second, separate, unexplained near-coincidence, and **not** the source of the $2/3$ (which holds for any δ).

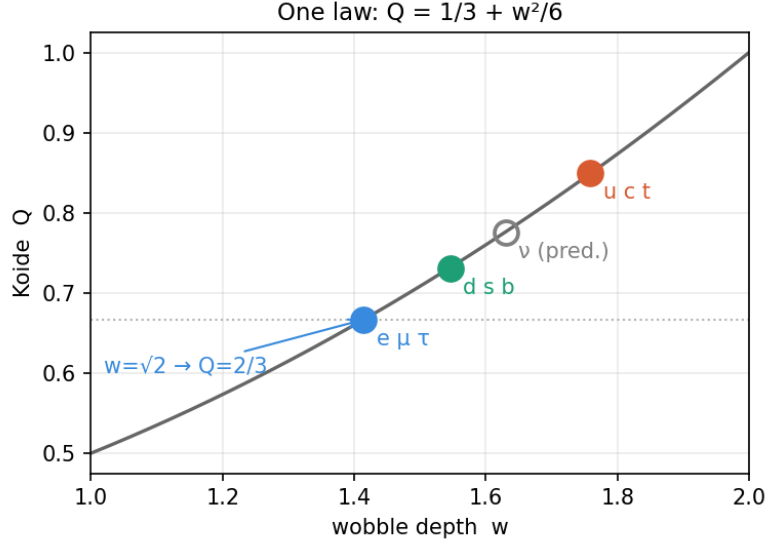


Figure 2: The single law $Q = 1/3 + w^2/6$. The charged leptons sit at $w = \sqrt{2}$ ($Q = 2/3$); the quark families at larger w ; the open circle is the neutrino corner extrapolated from the toy maps of §4.

3. The Event Density re-description (a shape-match, not a derivation)

Koide's skeleton is $(\sum m)/(\sum \sqrt{m})^2 = (\text{sum of squared amplitudes})/(\text{coherent sum, squared})$. That is the **amplitude-vs-probability structure of quantum mechanics**: add amplitudes and square to get interference; sum the squares to get probabilities. Event Density is built on exactly this object. Its participation carrier is $P = \sqrt{b} e^{i\pi_K}$ (π_K the polarity phase), its Born rule reads $f = |P|^2$ (Paper_003), and mass is **rest-frame bandwidth content b** (Paper_113, tier M2). Hence $\sqrt{m}$ maps onto the **ED amplitude of a mass-carrying mode** — if the three generations are read as coherent amplitude-channels, an organizing assumption rather than a corpus result — and Koide's Q is then a **coherence / spread ratio among the three generation-amplitudes** — how democratically the three lepton modes share amplitude — with $2/3$ a balanced value. Brannen's form then reads directly in ED terms: three amplitudes at 120° relative phase (Z_3), wobble depth $\sqrt{2}$, offset $\delta \approx 2/9$.

Tier. This is RE-DESCRIPTION. It re-expresses the *shape* of Koide (the roots, the square, the $1/3$ -to-1 range) in ED's own idiom, and nothing more; the Z_3 three-generation structure is an inherited *input* (Brannen's ansatz fitted to the three known generations), not something ED explains. It does not derive the masses, the generation count, $w = \sqrt{2}$, or $\delta \approx 2/9$. Its value is that it relocates the mystery into ED's native variables (amplitude, phase, coherence), which is where any future derivation would have to live — and coherence is the object ED's Σ functional and V5 budget already deal in.

4. Two toy maps over the real quantum numbers, and one prediction

Mass is not fundamental in the Standard Model; charge and weak isospin/hypercharge are, with mass downstream. So we ask whether the wobble w organizes over the *real* quantum numbers.

Charge (1-D). Against electric charge, the three charged families give $(-1, 1.414)$, $(-1/3, 1.546)$, $(+2/3, 1.759)$: per unit charge the successive slopes are $+0.20$ and $+0.21$ — nearly equal, so **w is approximately linear in charge, $w \approx 1.61 + 0.20 \cdot Q_{\text{charge}}$** . Put charge on a vertical axis and the wobble grows linearly with height — a straight-sided cone (Figure 3).

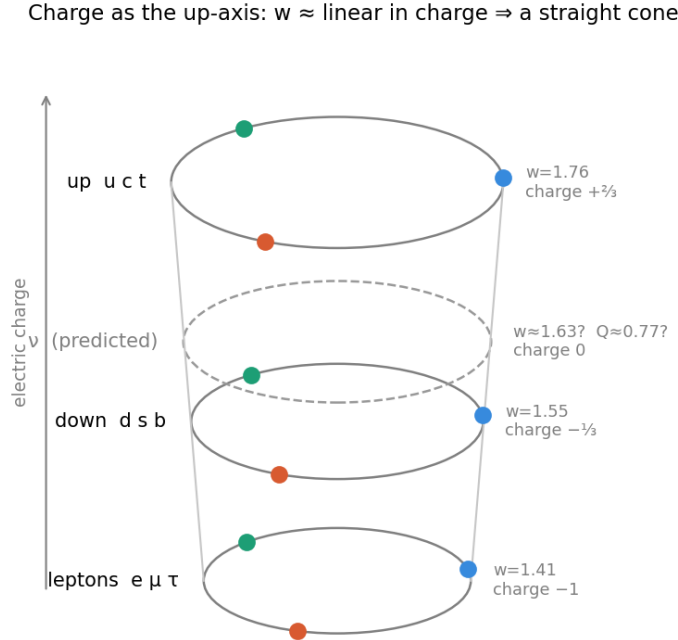


Figure 3: Charge as the up-axis. Because the wobble grows about linearly with electric charge, the four sectors stack into a straight-sided cone: leptons at charge -1 (bottom), up-quarks at $+2/3$ (top), each solid ring carrying its three generations. The dashed ring is the predicted neutrino corner at charge 0 ($w \approx 1.63$, $Q \approx 0.77$). This is the 1-D toy map; it fits three charged points and extrapolates the fourth.

Isospin $\times$ hypercharge (2-D). More honestly, the fermions live on a 2×2 doublet grid: weak isospin $T_3 = \pm \frac{1}{2}$ across (up-member vs down-member), hypercharge Y up (quark doublet $+\frac{1}{3}$, lepton doublet -1). Placing a Koide triangle at each site (Figure 4), the wobble is **monotone in both directions** — up-members exceed down-members, quarks exceed leptons — and lies close to a plane over the grid, largest at the up-quark corner, smallest ($\sqrt{2}$) at the electron.

The one prediction. The plane through the three charged corners puts the neutrinos ($T_3 = +\frac{1}{2}$, $Y = -1$) at **$w \approx 1.63$, Koide $Q \approx 0.77$** — the dashed ring of Figure 3, the dashed corner of Figure 4, and the open point of Figure 2. The 1-D charge line gives the same value. Neutrino masses are far too poorly known to test this today, but it is a concrete, falsifiable-in-principle output.

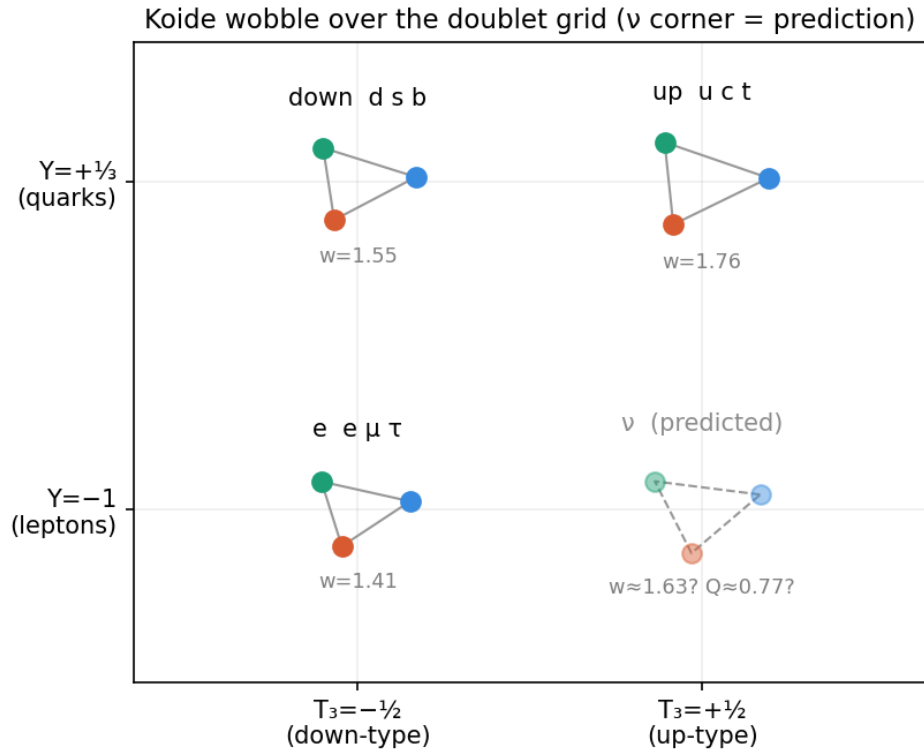


Figure 4: The four fermion types on their own Standard-Model coordinates (weak isospin across, hypercharge up), each a Koide triangle whose size is the wobble w . The wobble grows toward the up-quark corner. The dashed neutrino corner is a prediction, not data.

Tier — read this before believing the maps. These are TOY MODELS with soft legs. The lepton 2/3 (§1–2) is rigid and complete-data; the cross-family maps are not. The plane has three free parameters and there are exactly three charged data points, so it fits *automatically* and carries no residual to test — its only content is the neutrino corner. Two of the three inputs (the quark w’s) are scheme-fuzzy. And the Standard Model’s structure is not truly 1-D: up and down quarks are isospin partners, so a single clean axis can only be partial. So Figure 4 is a legitimate, non-arbitrary *map* (it uses the SM’s own labels, not invented ones) plus one filed prediction — not evidence of a law.

5. What connects to ED, and what does not

- **Compatible, at the level of structure.** ED grounds charge as integer topological winding (Paper_ChargeAsTopology_B4) and the SU(2)/doublet gauge structure from channel multiplicity, so the axes of Figure 4 are ED-adjacent, and every slice (a coherent superposition of three $\sqrt{m}$ amplitudes) is ED-native language.
- **Not derived.** ED does not produce the fractional hypercharges, the masses, the generation count, or the values $w = \sqrt{2}$, $\delta \approx 2/9$. So the docking port — the map’s axes and the reason the wobble takes the values it does — is exactly the part ED has not built.
- **The honest ED talking point (consilience-tier).** In the Standard Model, masses are arbitrary Yukawa dials, so Koide’s 2/3 is a fluke among free parameters. In ED, mass *is* bandwidth content and $\sqrt{m}$ its amplitude, so “the lepton masses fold into a clean amplitude-on-a-circle relation” is the *kind* of thing expected if masses are amplitudes and a surprise if they are dials. That is a modest point of consilience for ED’s ontology — not a derivation, and not to be inflated.

6. Bottom line

Read through Event Density, Koide’s relation is a **$\mathbf{Z}_3$ -symmetric, $\sqrt{2}$ -deep coherence pattern among three generation-amplitudes**, governed entirely by $Q = 1/3 + w^2/6$, with one unexplained phase ($\delta \approx 2/9$). Over the fermions’ real quantum numbers the wobble lays out as a smooth field — a legitimate map, with a single filed prediction (neutrino $Q \approx 0.77$). ED can say this much: it re-describes the structure in its own amplitude/coherence language and offers an honest toy map. It cannot derive the value, the generation count, or any mass — those are the Standard Model flavor sector, which ED inherits. The two things worth remembering are the **complete-data weight of the lepton 2/3** and the **neutrino prediction**, either of which becomes actionable only when neutrino masses sharpen or when ED’s gauge-and-generation sector is derived rather than inherited.